

Hands-on Exploration of Hybrid 4D Printing Design Space

Himani Deshpande
hdeshpande11@tamu.edu
Texas A&M University
USA

Courtney Starrett
cstarrett@tamu.edu
Texas A&M University
USA

Jinsil Hwaryoung Seo
hwaryoung@tamu.edu
Texas A&M University
USA

Clement Zheng
clement.zheng@gmail.com
National University of Singapore
Singapore

Jeeun Kim
jeeun.kim@tamu.edu
Texas A&M University
USA

ABSTRACT

4D printing is an emerging field of fabrication garnering interest from the HCI community in its potential applications. In this work we present a hands-on exploration of the hybrid design space for 4D printing which makers can utilize to create their own artifacts by designing parts to control their bend, supply different heat triggers, and integrate craft techniques based on the intended use.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

4D Printing, FDM, craft, makerspaces

ACM Reference Format:

Himani Deshpande, Courtney Starrett, Jinsil Hwaryoung Seo, Clement Zheng, and Jeeun Kim. 2022. Hands-on Exploration of Hybrid 4D Printing Design Space. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH '22 Labs)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532725.3535587>

1 INTRODUCTION

4D printing is an increasingly popular technique of fabrication where the 3D printed parts change shape over time when supplied with an external stimulus such as heat. Researchers have explored 4D printing in the fields of robotics [Chen et al. 2018; Etherington 2016], smart packaging [Kerry and Butler 2008], assembling furniture [An et al. 2018], artificial muscles [Takashima et al. 2010]. With the lowering cost of Fused Deposition Modeling (FDM) 3D printers and the off-the-shelf availability of a wide variety of materials [Simplify3D 2019] including materials showcasing 4D behavior such as PolyLactic Acid (PLA), we see an opportunity to translate the 4D printing knowledge from research labs to makerspaces, providing a new avenue for makers to explore creatively.

The research into the manipulation of printing parameters to affect the shape changing behavior such as changing printing temperature, speed, printing direction, layer height, etc. [An et al. 2018; Gu et al. 2019; Wang et al. 2019] and even using multimaterial printing to create composites with constraining layers [An et al. 2018; Qin et al. 2021; Tahouni et al. 2020] makes it possible for makers to integrate the knowledge in the making process. Venturing further into accessible techniques, hybrid fabrication can be helpful in implementing digital fabrication with hands-on and familiar craft techniques where makers can draw insights from their previous experiences and apply to the new 4D printing technology. In this work we present the hybrid fabrication and craft design space of 4D printing and the primitives that can be explored by makers in real time to bridge the gap between understanding of and designing for the technology.

2 HYBRID DESIGN SPACE

4D behaviour can be characterized in two ways: (1) Shape Memory Phenomenon, and (2) Programmable Shape change. Shape memory is the property of a material, PolyLactic Acid (PLA) in our case, to revert to the shape it is printed in after deformation on application of an external stimulus, for example, heat. It is also possible to manipulate printing parameters such as printing direction or raster angles, layer thickness, printing temperature, etc. to cause shape change directly after printing on application of a trigger without prior deformation. This is considered programmable shape change.

In the 4D hybrid design space [Deshpande et al. 2022] (see Figure 1), we present ways in which makers can affect the shape change of the printed material through bend control techniques of digital fabrication and manipulating printing parameters, using multi-material printing to add constraining layer to ensure bend direction, and manual intervention of cutting and shaping the printed parts to affect their resultant shape change. We also showcase and categorize the different heat triggers depending on whether the heat source is attached or embedded such as a resistive heating element or if the heat sources is external, heating the material globally or locally. Through the craft integration, we want to show the opportunity of combining familiar techniques and materials with 4D printed parts. Using the design space, we showcase four primitives (see Figure 2) created by choosing different elements of the design space. Users can globally change shape of printed parts using hot water, embed heating element such as conductive thread or nichrome wire in designed channels to shape parts locally or globally, attach printed parts to fabric with sewn conductive thread

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Labs, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9367-6/22/08.

<https://doi.org/10.1145/3532725.3535587>

to add textures to the fabric, and even create hinges by embedding resistive elements in designed channels.

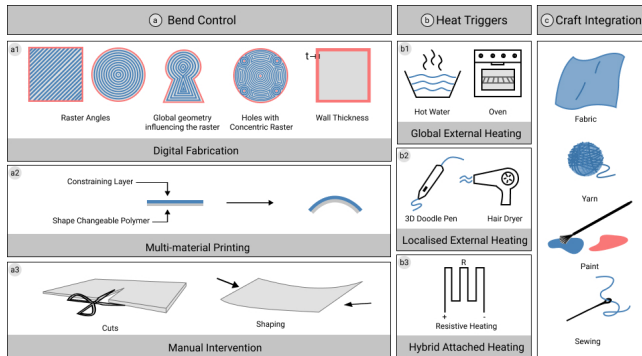


Figure 1: Hybrid 4D Design Space [Deshpande et al. 2022]: (a) Bend control techniques, (b) Heat triggers, and (c) Craft integration

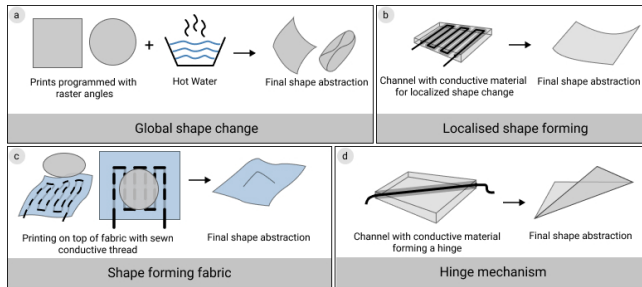


Figure 2: Examples of primitives [Deshpande et al. 2022]: (a) Global shape change, (b) Localised shape forming, (c) Shape forming fabric, and (d) Hinge mechanism

3 HANDS-ON EXPLORATION

As part of the hands-on exploration, attendees can understand about the 4D printing technology through a tutorial. To show the use of the design space and primitives, we will display inspiring examples of applications that we have developed such as scaled example of collapsible multi-purpose furniture, and some jewelry (see Figure 3). Attendees will have a chance to explore the pre-printed parts and primitives to think of ideas for using 4D printing. We will set up a design and printing station for interested attendees to print their desired designs, either from a predefined library or their own based on their knowledge of 3D modeling. We will also provide pre-printed primitives with channels for attendees to explore embedded heat triggers as well as provide craft material for them to explore the shape change through cutting, or make additions to their explorations with fabric, paint, etc.

REFERENCES

Byoungkwon An, Ye Tao, Jianzhe Gu, Tingyu Cheng, Xiang 'Anthony' Chen, Xiaoxiao Zhang, Wei Zhao, Youngwook Do, Shigeo Takahashi, Hsiang-Yun Wu, Teng Zhang, and Lining Yao. 2018. Thermorph: Democratizing 4D Printing of Self-Folding Materials and Interfaces. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173834>

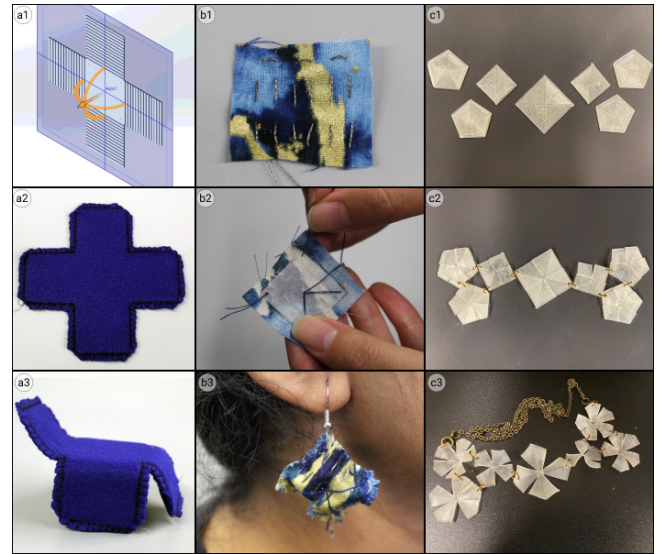


Figure 3: Application examples [Deshpande et al. 2022]: (a1-a3) Scaled model of collapsible multi-purpose seating, (b1-b3) Fabric earrings, and (c1-c3) Necklace

- Tian Chen, Osama R Bilal, Kristina Shea, and Chiara Daraio. 2018. Harnessing bistability for directional propulsion of soft, untethered robots. *Proceedings of the National Academy of Sciences* 115, 22 (2018), 5698–5702.
- Himani Deshpande, Clement Zheng, Courtney Starrett, Jinsil Hwaryoung Seo, and Jeeun Kim. 2022. Fab4D: An Accessible Hybrid Approach for Programmable Shaping and Shape Changing Artifacts. In *Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Daejeon, Republic of Korea) (TEI '22). Association for Computing Machinery, New York, NY, USA, Article 70, 7 pages. <https://doi.org/10.1145/3490149.3505574>
- Darrell Etherington. 2016. MIT's shape-shifting 3D-printed objects could improve medicine and solar-power. <https://techcrunch.com/2016/08/26/mit-shape-shifting-3d-printed-objects-could-improve-medicine-and-solar-power/> (Accessed on 1/16/2022).
- Jianzhe Gu, David E. Breen, Jenny Hu, Lifeng Zhu, Ye Tao, Tyson Van de Zande, Guanyun Wang, Yongjie Jessica Zhang, and Lining Yao. 2019. Geodesy: Self-Rising 2.5D Tiles by Printing along 2D Geodesic Closed Path. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3290605.3300267>
- Joseph Kerry and Paul Butler. 2008. *Smart packaging technologies for fast moving consumer goods*. John Wiley & Sons.
- Fang Qin, Huai-Yu Cheng, Rachel Sneeringer, Maria Vlachostergiou, Sampada Acharya, Haolin Liu, Carmel Majidi, Mohammad Islam, and Lining Yao. 2021. ExoForm: Shape Memory and Self-Fusing Semi-Rigid Wearables. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 249, 8 pages. <https://doi.org/10.1145/3411763.3451818>
- Simplify3D. 2019. Properties Table. <https://www.simplify3d.com/support/materials-guide/properties-table/> (Accessed on 10/17/2021).
- Yasaman Tahouni, Tiffany Cheng, Dylan Wood, Renate Sachse, Rebecca Thierier, Manfred Bischoff, and Achim Menges. 2020. Self-Shaping Curved Folding: A 4D-Printing Method for Fabrication of Self-Folding Curved Crease Structures. In *Symposium on Computational Fabrication* (Virtual Event, USA) (SCF '20). Association for Computing Machinery, New York, NY, USA, Article 5, 11 pages. <https://doi.org/10.1145/3424630.3425416>
- Kazuto Takashima, Jonathan Rossiter, and Toshiharu Mukai. 2010. McKibben artificial muscle using shape-memory polymer. *Sensors and Actuators A: Physical* 164, 1-2 (2010), 116–124.
- Guanyun Wang, Ye Tao, Ozguc Bertug Capunaman, Humphrey Yang, and Lining Yao. 2019. A-Line: 4D Printing Morphing Linear Composite Structures. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300656>